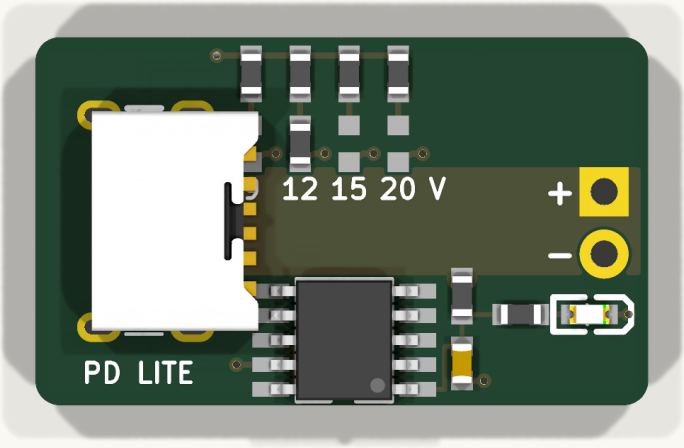
pd-trigger-lite

A 25 by 15 mm USB-C PD trigger: plug in a PD charger and get 9, 12, 15 or 20 V at up to 3 A on two solder pads.

This board asks a USB-C Power Delivery charger for one fixed voltage and passes the charger's VBUS straight to two 2.54 mm output holes. There is no regulator and no switch in the power path, so the output is exactly what the charger delivers. The voltage is picked by one small solder link, and the board ships set to 12 V. It is meant for bench use: powering a strip, a router or a prototype from a laptop charger.



TOP VIEW 25 X 15 MM, 2 LAYERS, 1 OZ RENDER

USB-C receptacle on the left, the voltage links along the top, the PD controller at the bottom, output pads on the right (+ is VBUS, - is ground). The red LED sits beside the output pads.

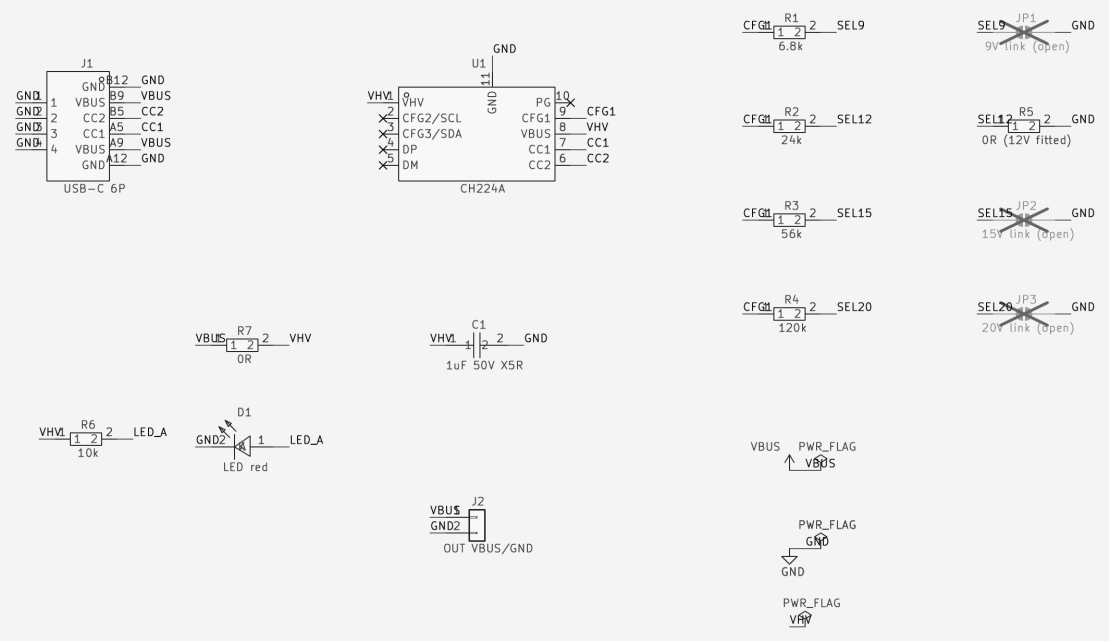
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Schematic

One controller, one capacitor, four resistors and a link. Everything else is copper.

The receptacle J1 is a 6-pin power-only part: two VBUS contacts, two ground contacts and the two CC lines. The controller U1 (WCH CH224A) talks to the charger over CC1 and CC2 and needs nothing else to negotiate. It is powered from VBUS through the 0 ohm link R7, which only exists so the controller's pins can be reached with thin copper, and C1 (1 uF, 50 V) decouples it. VBUS runs as a wide copper pour from J1 to the output pad J2.

U1 reads the resistance from its CFG1 pin to ground once, at power-up, and requests the matching voltage: 6.8k gives 9 V, 24k gives 12 V, 56k gives 15 V and 120k gives 20 V. Each of those resistors (R1 to R4) reaches ground only through its own link, so the one closed link decides the voltage. R6 and D1 light the LED from VBUS.



SCHEMATIC KICAD/PD-TRIGGER-LITE.KICAD_SCH ERC 0 ERRORS, 0 WARNINGS

Nets join by label name. JP1 to JP3 are empty 0603 lands; R5 is the 0 ohm resistor fitted in the 12 V position.

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Using it

Set the voltage before the first plug-in, check it with a meter, then connect the load.

VOLTAGE	LINK	WHAT IS FITTED
9 V	JP1	R1 6.8k in series
12 V	R5	R2 24k in series; shipped fitted with a 0 ohm resistor
15 V	JP2	R3 56k in series
20 V	JP3	R4 120k in series

To change the voltage, unplug the board, remove the 0 ohm resistor R5 with the iron, and either solder it onto the land marked with the voltage you want or bridge that land with a blob of solder. The four links sit in a row under the resistor row, labelled 9, 12, 15 and 20 on the silkscreen.

EXACTLY ONE LINK

With no link closed, CFG1 floats and the request is undefined. With two closed, two resistors sit in parallel and the controller reads a value that is in no table. Close one link, never zero and never two.

Output. The two plated holes take 20 AWG wire or a 1x2 2.54 mm header, which you solder yourself; JLC does not fit them. The rating is 3 A at any voltage. A charger without an e-marked cable never offers more than 3 A, and the board makes no 5 A claim.

LED. D1 is lit whenever VBUS is present. Its current rises with the voltage, so it glows faintly at 5 V and clearly at 12 V and above. A dim LED after plugging in means the charger stayed at 5 V: it did not offer the voltage you picked, or no link is closed.

First power-up. Plug in with nothing on the output, measure the voltage across the pads, and only then connect the load. The charger must offer the chosen voltage as a fixed PDO; many phone chargers stop at 9 V. The board has no fuse and no TVS diode, because the controller’s datasheet asks for neither. A hot-plug spike at 20 V is the one plausible way to damage it, so at 20 V plug the charger into the wall first and the cable into the board last.

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What is on it

Eleven parts placed by JLC, two of them Extended. Everything else is bare copper.

REF	PART	LCSC	TYPE	QTY
U1	CH224A PD sink controller, ESSOP-10	C42459160	Extended	1
J1	USB-C receptacle 6P, 3 A 20 V (TYPE-C-6M-001)	C2798175	Extended	1
C1	1 uF 50 V X5R 0603	C15849	Basic	1
R1	6.8k 1% 0603	C23212	Basic	1
R2	24k 1% 0603	C23352	Basic	1
R3	56k 1% 0603	C23206	Basic	1
R4	120k 1% 0603	C25808	Basic	1
R5, R7	0 ohm 0603	C21189	Basic	2
R6	10k 1% 0603	C25804	Basic	1
D1	red LED 0603	C2286	Basic	1

Not assembled: JP1 to JP3 are empty 0603 lands (the 9, 15 and 20 V links), and J2 is the pair of output holes. The two Extended parts each add JLC's per-part feeder fee; no Basic part can replace either of them.

What it costs

About 27 USD for five assembled boards before shipping, as an estimate.

24.33
USD, JLC estimate for 5 boards: PCB 2.00, setup 8.00, 2 Extended fees 6.00, stencil 8.00, joints 0.33.

2.78
USD, parts for 5 boards at LCSC unit prices, about 0.56 per board.

27.1
USD total for 5 boards, before shipping and tax.

These are estimates from the skill's price table (verified 2026-07-24), not a quote. Live JLC prices have run 1.9 to 3.1 times higher than that table for bare PCBs, so expect the PCB

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line to be nearer 4 to 6 USD. Shipping depends on the country and the carrier and is not included. The cart at `https://cart.jlcpcb.com/quote` is the only real price.

Estimate from fab/quote.json (order_quote.py, qty 5, HASL, economic assembly) and kicad/parts.json prices, read 2026-09-24.

Ordering at JLCPCB

Four files go up: the gerber zip, then BOM.csv and CPL.csv for assembly.

1. Open the JLC quote page (link above) and upload the gerber zip, `fab/pd-trigger-lite_gerbers.zip`.
2. Check that the viewer reads 2 layers, 25 x 15 mm. Keep 1.6 mm, 1 oz, HASL, green, quantity 5.
3. Turn on PCB Assembly: Economic, top side, 5 boards.
4. Upload `fab/BOM.csv` and `fab/CPL.csv`. Confirm all eleven parts matched their LCSC numbers and that JP1, JP2, JP3 and J2 are absent, since those are not assembled.
5. In the placement preview, check three parts. D1: its cathode mark must face the right board edge. U1: the pin-1 dot sits at the corner nearest the capacitor C1. J1: the receptacle mouth points off the left edge. No rotation correction was applied to the CPL, because every footprint comes from JLC's own library.
6. Run JLCDFM on the gerbers as a second opinion, then pay. Payment is yours to do.

Gates on 2026-09-24: ERC 0/0, DRC 0 errors 0 unconnected (reports/drc_final.json), verify and DFM pass.

What it cost to design

About 24 USD of model spend went into designing this board. It is not part of the board's price.

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STEP	USD
Before the workspace (brief and skill read)	5.29
Brief and intake	2.92
Architecture	0.03
Schematic	5.21
Placement	0.51
Routing	2.24
Verification	5.38
DFM, order files and guide	2.67
Total	**24.26**

The split follows when hwde recorded each step, and the rows are rounded to the cent. The worker loop logged only 9.37 USD, because one of its four iterations timed out and a timed-out iteration logs no cost. The session transcripts hold the whole 24.26 USD. Later skill-wide rebuilds of this guide are skill work and are not counted.

reports/cost.json (gen_cost.py), priced from reference/model_prices.yaml, which was fitted to Claude Code's own session totals on 2026-09-24.

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